

Bhuman Pandita

AI Engineer | Applied AI | Agentic Systems | GenAI

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SUMMARY

AI Engineer with hands-on production experience building and deploying **multi-agent LLM systems**, **RAG pipelines**, and **agentic AI workflows** at scale. Proven track record of owning end-to-end AI system design, from architecture and experimentation through production deployment, stakeholder delivery, and measurable business impact. Strong foundation in **Azure OpenAI**, **Semantic Kernel**, **MCP servers**, **LangChain/LangGraph**, **FastAPI**, and async Python systems. Published researcher and ASCE award recipient.

EXPERIENCE

- Machine Learning Engineer (Assistant Manager), IndiGo (InterGlobe Aviation)** Jul 2026 – Present
- Promoted to **Machine Learning Engineer (Assistant Manager)** in Jul 2026.
- IAL - Digital & Data Science Program, IndiGo (InterGlobe Aviation)** Jun 2025 – Jul 2026
- Owned end-to-end design and **production deployment of a multi-agent MCP server** (Semantic Kernel, Azure OpenAI, FastAPI, FastMCP) automating BluChip loyalty profile updates across multiple internal databases — **reduced TAT from 12 days to under 1 minute** with PII compliance and OCR-based government ID verification, directly improving operational throughput for IndiGo's loyalty team.
 - Architected and delivered a **hybrid AI Loyalty Fraud Detection system** integrating MS Graph API, Azure Cosmos DB, 3 Capillary APIs, and Navitaire — 6 detection modules including **LLM-driven geographic feasibility reasoning (zero hardcoded thresholds)** and self-connecting flight protection; sole engineer on end-to-end POC delivered in 3 months with a pivot from microservices to async monolith for production reliability.
 - Prototyped an **Edge AI IoT asset tracker** using a custom CNN vision model on Raspberry Pi with Go + Python backend, validating embedded AI inference for digitizing non-digital airline ground assets — presented to senior engineering leadership.
 - In progress:* **NLP/LLM invoice categorization pipeline** (embedding-retrieval + LLM classification) to auto-tag MRO Non-Routine Card tasks across heterogeneous vendor invoices, targeting lessor compliance and audit automation.
- AI Engineer Intern, Swostitech Solutions** Dec 2024 – Jan 2025
- Built and optimized a **production RAG pipeline** over 1,000+ enterprise documents — benchmarked OpenAI GPT-4, LLaMA-3, and Mistral-7B across chunking strategies and embedding models, achieving measurable gains in answer relevance and retrieval grounding.
 - Reduced end-to-end RAG inference latency by 30% (p95)** via embedding model quantization, flat-to-HNSW vector store migration, and query-layer caching — enabling higher concurrency without hardware scaling.
- Data Analyst Intern, Rigi, Bangalore** Jul 2024 – Dec 2024
- Automated end-to-end transaction reporting for 100+ creators (pandas, openpyxl, cron), **saving ~20 finance-team hours/week (50% effort reduction)** and surfacing payout discrepancies automatically — eliminating a previously manual weekly process.
 - Optimized PostgreSQL queries (window functions, CTEs, index tuning) **cutting Metabase dashboard load time by 30%**, directly improving analyst productivity and creator-economy insight delivery.

PROJECTS

- RAG Red-Teaming Pipeline** | *Python, Gemini, Groq/LLaMA-3.3-70B, FastAPI, Next.js* 2025
- Designed and built a **4-stage adversarial AI evaluation framework** (attacker → executor → target RAG → evaluator) to systematically probe an HR chatbot across **13 attack categories** — prompt injection, roleplay jailbreak, encoding obfuscation, RAG context poisoning, and more.
 - Implemented a **hybrid AI observability scoring system** (0.7× LLM semantic judge + 0.3× heuristic signals, CRITICAL/HIGH/MEDIUM severity tiers) with real-time SSE-streaming campaign dashboard in Next.js.
- CycleGAN Image Dehazing** | *PyTorch, OpenCV, Deep Learning* Mar – May 2024
- Designed and trained a **CycleGAN-based image dehazing model** from scratch — unpaired hazy-to-clear image translation using cycle consistency loss and adversarial training, achieving **40% improvement in perceptual clarity** across diverse real-world haze conditions.
- CNN Sound Classification** | *PyTorch, Librosa, Mel spectrograms* Jan – Mar 2024
- Benchmarked ResNet, DenseNet, MobileNetV2 and a custom 6-layer CNN on **6,000+ audio samples** across 13 classes; noise reduction preprocessing on Mel spectrograms **improved classification accuracy by 15%**.

EDUCATION

Birla Institute of Technology and Science, Pilani

Pilani, India

*B.E. Civil (Hons.) with **Minor in Data Science**, CGPA 7.8/10*

Aug 2021 – May 2025

Relevant Coursework: Generative AI, Deep Learning, Machine Learning, Foundations of Data Science, Applied Statistical Methods.

PUBLICATIONS & AWARDS

- Nanavaty, A., Sharma, R., **Pandita, B.** et al. *Integrating deep learning for VQA in Agricultural Disease Diagnostics: Case Study of Wheat Rust*. **Sci Rep** 14, 28203 (2024). doi:10.1038/s41598-024-79793-2
- Rodríguez-Narvaez, O. M., **Pandita, B.**, Goyal, O., et al. *Efficacy of biochar as a catalyst for a Fenton-like reaction*. **J. Water Process Eng.**, 70, 107014 (2025).
- Morales-Verdin, B.S., Goyal, O., **Pandita, B.**, et al. *Optimizing the upcycling of microplastics to a carbon-based adsorbent for water treatment: An integrated experimental and computational approach*. **Chem. Eng. Sci.** 325, 123409 (2026). doi:10.1016/j.ces.2026.123409
- **First Prize**, ASCE International Conference (CISSC-2025) — UG Event on Challenges and Innovations for Sustainable Smart Cities, ASCE India Section.

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL), R, Go (basic), Spark (PySpark)

LLM & Agentic AI: Semantic Kernel, LangChain, LangGraph, AutoGen, FastMCP, MCP Servers, Multi-Agent Orchestration, RAG, Prompt Engineering, Function Calling, Structured Outputs, Guardrails, AI Observability

Models & APIs: Azure OpenAI, OpenAI, Google Gemini, LLaMA-3, Mistral, Groq, Embedding Models

ML / DL / Statistics: PyTorch, TensorFlow, scikit-learn, XGBoost, CNNs, GANs, CycleGAN, Computer Vision, NLP, Statistical Modelling

Data & Engineering: pandas, NumPy, FastAPI, asyncio, Pydantic, Docker, Git (GitHub), Azure Cosmos DB, MS Graph API, SQLite, Streamlit, Jira

BI & Visualization: Tableau, Power BI, Metabase, Postman, Bruno

Cloud & Infrastructure: Azure OpenAI, AWS (familiar), Docker, production pipeline deployment & maintenance